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Tree Reconstruction from Metric Distances and Additive Metrics

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Mathematical Treatise Summary (verbatim)

A distance metric D on a finite set of taxa X is additive if and only if there exists a weighted tree T with leaf set X whose path lengths exactly realize D. Additive metrics satisfy the four-point condition: for any four leaves, the two largest sums of pairwise path distances are identical. We analyze exact and approximate tree reconstruction algorithms, demonstrating that additive distance matrices contain sufficient mathematical information to uniquely determine both the unobserved internal branching topology and all edge lengths.

Semple & Steel (2003), Oxford University Press

Notes: record the page or section for every claim you make about this paper.